

Product Requirements Document (PRD)

Learning Management System (LMS)

Version: 1.1

1. Product Overview

Objective

Develop a Learning Management System (LMS) for coaching institutes that enables administrators to manage academic classes, subjects, subject-wise pricing, student enrollments, batches, recurring sessions, and attendance.

The system is designed for institutes where students can enroll in multiple subjects and belong to different batches for each subject. Teachers manage only their assigned batches, while students can track their own attendance.

2. User Roles

Super Admin

The Super Admin manages the complete academic structure and user administration.

Responsibilities

- Create and manage Staff accounts
- Create and manage Student accounts
- Configure Academic Years
- Configure Classes (8–12)
- Create Subjects
- Configure Subject Offerings
- Define subject fees per class
- Create and manage Batches
- Assign Staff to Batches
- Enroll Students into Subject-specific Batches
- Create recurring Session Templates
- View and edit Attendance
- Generate Reports
- Transfer Students between batches

- Activate/Deactivate Staff and Students
-

Staff (Teacher)

Teachers manage only the batches assigned to them.

Responsibilities

- View assigned batches
- View today's sessions
- Mark attendance
- View attendance history
- View student attendance percentage
- View enrolled students in their batches

Teachers cannot modify batches, enrollments, or pricing.

Student

Students have read-only access.

Responsibilities

- View enrolled subjects
 - View assigned batches
 - View attendance history
 - View attendance percentage
-

3. Core Entities

Academic Year

Represents an academic cycle.

Example:

2026–2027

Used for maintaining historical records.

Academic Class

Represents the student's grade.

Examples

- Class 8
 - Class 9
 - Class 10
 - Class 11
 - Class 12
-

Subject

A reusable master record.

Examples

- Mathematics
- Science
- Physics
- Chemistry
- Biology
- English

Subjects are independent of classes.

Subject Offering

A Subject Offering links a Subject to a Class.

It also stores the fee charged for that subject in that class.

Example

Class	Subject	Fee
8	Science	₹2,500
9	Science	₹3,000
10	Science	₹3,500
11	Physics	₹5,000

Class	Subject	Fee
12	Physics	₹5,500

This design supports different pricing for different classes without duplicating subject records.

Batch

A Batch is created for a Subject Offering.

Each Batch contains:

- Batch Name
- Subject Offering
- Assigned Teacher
- Capacity (Optional)
- Status

Example

Batch A

Class 11 Physics

Teacher: Mr. Arun

Student Enrollment

Students are enrolled into batches instead of being permanently assigned.

Each Enrollment contains:

- Student
- Academic Year
- Subject Offering
- Batch

Example

Student

Physics → Batch A

Mathematics → Batch C

Chemistry → Batch B

This allows students to attend different batches for different subjects.

Session Template

A Session Template defines the recurring schedule of a batch.

Each template contains:

- Batch
- Subject
- Teacher
- Day of Week
- Start Time
- End Time

Example

Monday

5:00 PM – 6:00 PM

Physics

Batch A

Teacher: Mr. Arun

Session templates are reused throughout the academic year.

Attendance

Attendance is recorded against every scheduled occurrence of a session.

Each record contains

- Session
- Date
- Student
- Attendance Status
- Marked By
- Timestamp

Attendance history should remain unchanged after submission unless edited by the Super Admin.

4. Academic Structure

Academic Year



Academic Class



Subject Offering



Batch



Session Template



Attendance

Students are connected through Enrollment.

Student



Enrollment



Batch

5. Business Rules

Subject Availability

Subjects available depend on the student's academic class.

Example

Classes 8–10

- Science
- Mathematics
- English

Classes 11–12

- Physics
 - Chemistry
 - Biology
 - Mathematics
-

Multiple Subject Enrollment

A student may enroll in one or more subjects.

Example

Physics

Chemistry

Mathematics

Each enrollment can belong to a different batch.

Batch Assignment

Students are assigned by the Super Admin.

Teachers cannot move students between batches.

Subject Fees

Fees are maintained at the Subject Offering level.

Fee = Class + Subject

Not Subject alone.

Attendance

Attendance is recorded per scheduled session.

Attendance cannot exist without a scheduled session.

Batch Transfers

Students may change batches during the academic year.

Historical attendance remains linked to previous batches.

Session Scheduling

Sessions are recurring schedules.

Changes are expected to be rare.

Future sessions automatically follow the updated template.

6. Functional Modules

Academic Management

- Academic Years
 - Classes
 - Subjects
 - Subject Offerings
 - Subject Fees
-

User Management

- Staff
 - Students
-

Batch Management

- Create Batch
 - Assign Teacher
 - Configure Capacity
 - Enable/Disable Batch
-

Enrollment Management

- Assign Student
 - Change Batch
 - Remove Enrollment
 - View Enrollment History
-

Session Management

- Create Session Templates
 - Modify Schedule
 - Cancel Session (Optional)
-

Attendance

- Daily Attendance
 - Session Attendance
 - Student Attendance
 - Attendance History
 - Attendance Percentage
-

Reports

- Student Attendance Report
- Batch Report
- Subject Report
- Teacher Report

- Daily Attendance
- Monthly Attendance
- Subject Enrollment Report
- Fee Summary

7. Permissions Matrix

Feature	Admin	Staff	Student
Manage Classes	✓	✗	✗
Manage Subjects	✓	✗	✗
Configure Subject Fees	✓	✗	✗
Create Batches	✓	✗	✗
Assign Teachers	✓	✗	✗
Enroll Students	✓	✗	✗
View Assigned Batches	✓	✓	✓
Create Sessions	✓	✗	✗
Mark Attendance	✓	✓	✗
View Attendance	✓	✓	✓
Transfer Students	✓	✗	✗
Generate Reports	✓	✓ (Own Batches)	✗

8. Non-Functional Requirements

- Role-based access control.
- Audit logs for attendance modifications.
- Attendance submission locking.
- Support for future fee revisions.
- Scalable data model supporting additional classes and subjects.
- Historical academic records retained across years.

9. Future Scope

- Homework Management
- Online Tests & Exams
- Study Materials
- Parent Portal
- Online Fee Payments
- Student Performance Analytics
- Push Notifications
- Leave Management
- Timetable Calendar Integration